

kernel and the kernel is the one responsible for sending that message out to the speakers.

So the kernel then is like a middleman. Why does your device need a middleman, you ask? It helps keep the pressure away from Android to create operating systems that are capable of being compatible with all the different devices that they work on. The onus to create kernels lies on the manufacturers. The kernel is designed by the manufacturers for their own device, to work with the specs of that particular model. This leaves Android to work on making their operating system as flexible as they would like it to be.

Customized kernels

The manufacturer will need to have a number of drivers that they will need to plug in as well as a lot of variables that they need to adjust when it comes to creating a kernel. Which leads to added flexibility for creating a custom kernel for their device. They can change the following -

- The frequencies of the CPU - scale them up or down as needed.
- Adjust the voltage of the CPU regardless of the frequency chosen .
- The level at which the CPU engages the extra cores that it disables to save battery.
- Allow for USB fast charge through a 3.0 port.
- Maximum possible frequency for the GPU (Graphics Processing Unit).

So why choose a customized kernel? For the simple reason that you can choose a kernel that can provide your device with a boost that you need. Certain kernels will help better your device's speed and performance while others might help give your battery a boost. You could even choose a kernel that helps you better both. Developers have also had success with designing kernels using features from other kernels that do not officially exist for your device. Getting the right kernel for your device will take you a long way to building the exact custom ROM that you've envisioned so far.

Finding Android kernels

Finding a kernel for your Android device is a comparatively easy job. All you have to do is visit the XDA forums website and do a search for kernels compatible with your device.

Before you go shopping on XDA, make sure you have a rooted device as well as a custom recovery installed on your device. Once you have found the kernel of your choice, download the .zip file for it and flash it onto your